



TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC 2 2018

CANDIDATE
NAME

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CENTRE
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INDEX
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CLASS

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H2 BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

Thursday 13 September 2018
2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Section A

Write your name, Centre Number, index number and class in the spaces at the top of the page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graph.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the separate Writing Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Q1	/14
Q2	/13
Q3	/23
Q4 / 5	/25
Total	/75

This document consists of **21** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

- 1** Scientists investigated three genes, **C**, **D** and **E**, involved in controlling cell division. They studied the effect of mutations in these genes on the risk of developing lung cancer.

The scientists analysed genes **C**, **D** and **E** from healthy people and people with lung cancer.

- If a person had a normal allele for a gene, they used the symbol **N**.
- If a person had two mutant alleles for a gene, they used the symbol **M**.

They used their data to calculate the risk of developing lung cancer for people with different combinations of **N** and **M** alleles of the genes. A risk value of 1.00 indicates no increased risk.

Table 1.1 shows the scientists' results.

Table 1.1

Gene C	Gene D	Gene E	Risk of developing lung cancer
N	N	N	1.00
M	N	N	1.30
N	N	M	1.78
N	M	N	1.45

N = at least one copy of the normal allele is present

M = two copies of the mutant allele are present

- (a)** Suggest the relative importance of the mutant alleles of genes **C**, **D** and **E** on the risk of developing lung cancer. Explain your answer.

[3]

Chemotherapy is the use of a drugs, such as vinblastine, to treat cancer. The drug kills dividing cells. Fig. 1.1 shows the number of healthy cells and cancer cells in the blood of a patient receiving chemotherapy. The arrows labelled F to I show when the drug was given to the patient.

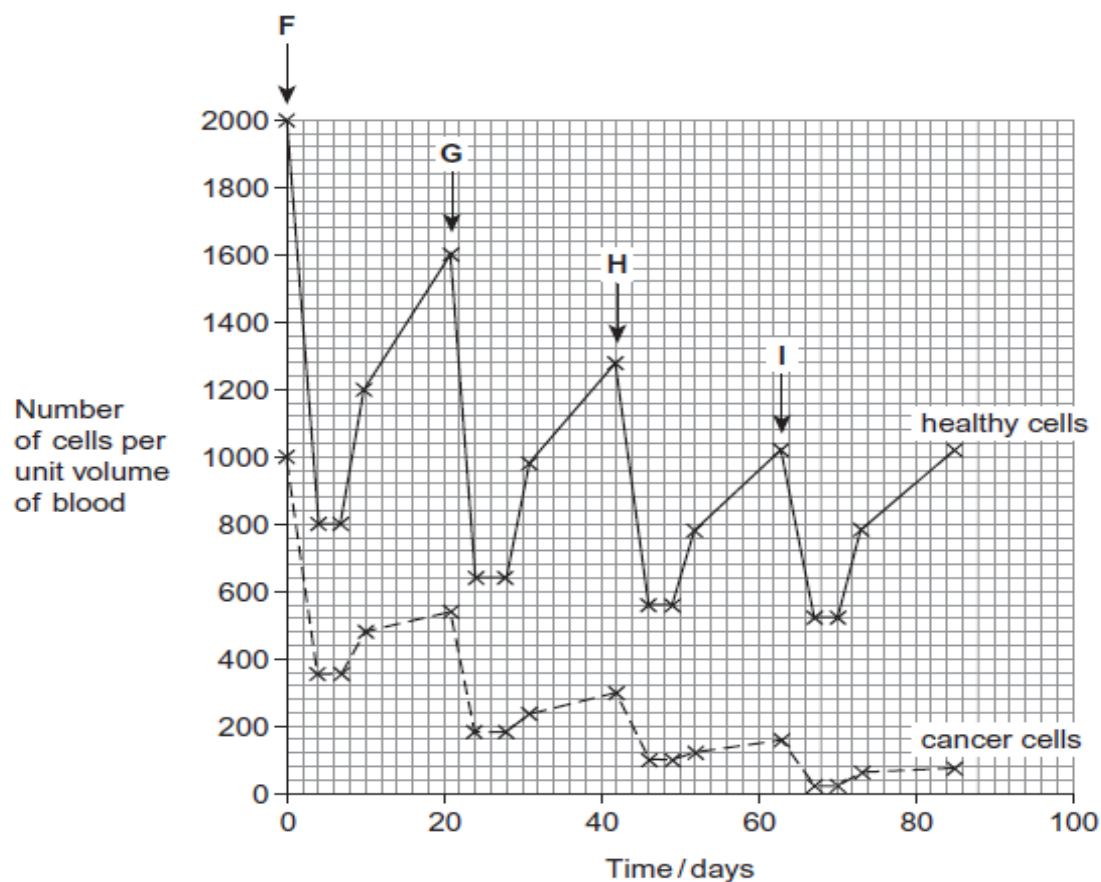


Fig. 1.1

- (b) Calculate the rate at which healthy cells were killed between days 42 and 46.

_____ cells killed per unit volume of blood per day

[1]

- (c) Describe **two** similarities and **one** difference in the response of healthy cells and cancer cells to the drug between times **F** and **G**.

[3]

- (d) More cancer cells could be destroyed if the drug was given more frequently. Suggest why the drug was **not** given more frequently.

[2]

- (e) State **two** ways in which cancer cells differ from normal healthy cells.

[2]

Vinblastine disrupts the formation of the spindle apparatus during mitosis.

(f) Explain how vinblastine exerts its effect as an anti-cancer drug.

[3]

[Total: 14]

- 2 The area over which the Arctic ice sheet extends varies throughout the year. Fig. 2.1 shows the variation in the extent of the Arctic ice sheet for the months of July to November for the years 1979 and 2009.

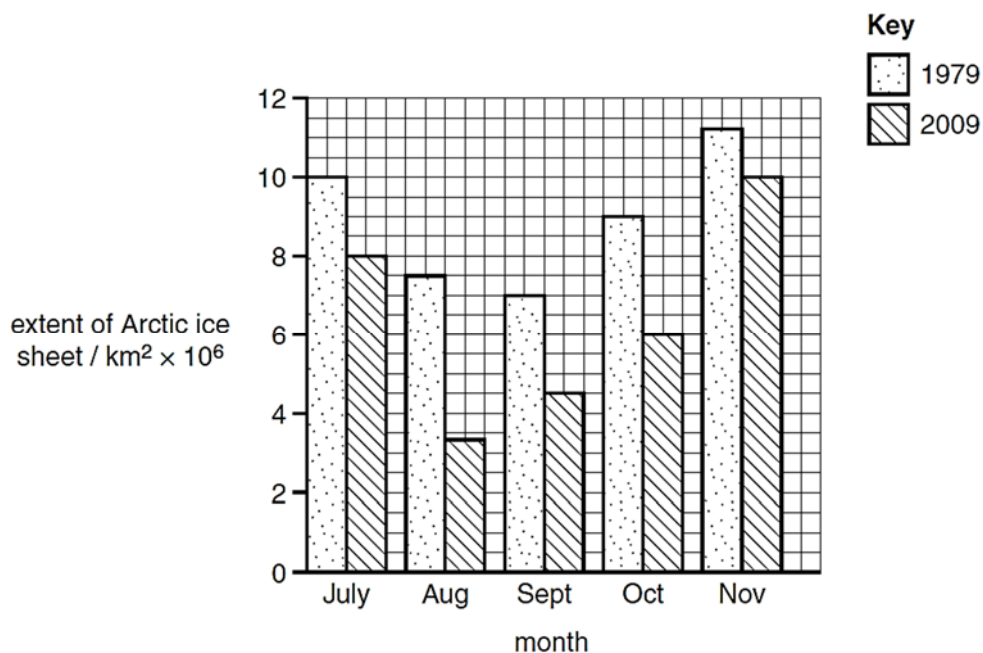


Fig. 2.1

- (a) Calculate the percentage reduction in the area over which the ice sheet extends between 1979 and 2009 for the month of September.

[1]

- (b) Suggest reasons for the reduction in the Arctic ice sheets from 1979 to 2009.

[2]

- (c) The polar bear, *Ursus maritimus*, moves across the Arctic ice sheet to hunt prey such as seals. When seals surface to breathe at cone-shaped breathing holes on the sea ice, a hunting polar bear which is waiting by the breathing hole will smack the head of the seal with both of its front paws to stun it, before biting and dragging the seal onto the ice. This method of still-hunting minimizes energy consumption and is the most successful strategy of hunting.

In 2008 the government of the USA classified *U. maritimus* as an endangered species because it is under the threat of extinction.

Suggest how climate change could have caused *U. maritimus* to become an endangered species.

[2]

Climate change also affects plants.

Plants can be categorized based on the way they photosynthesize. Most plants are C₃ plants because their first photosynthetic product is a three carbon compound. Examples of C₃ plants include barley, oats, potato, rice, and wheat commonly grown in temperate regions.

On the other hand, C₄ plants produce a four-carbon compound as their first photosynthetic product. Examples of C₄ plants are common grass crops of tropical regions, such as maize, millet, sorghum and sugarcane.

The rate of carbon dioxide uptake at a range of carbon dioxide concentrations by barley, a C₃ plant, and sugar cane, a C₄ plant, were compared at two temperatures.

The results of the experiment are presented in Fig. 2.2.

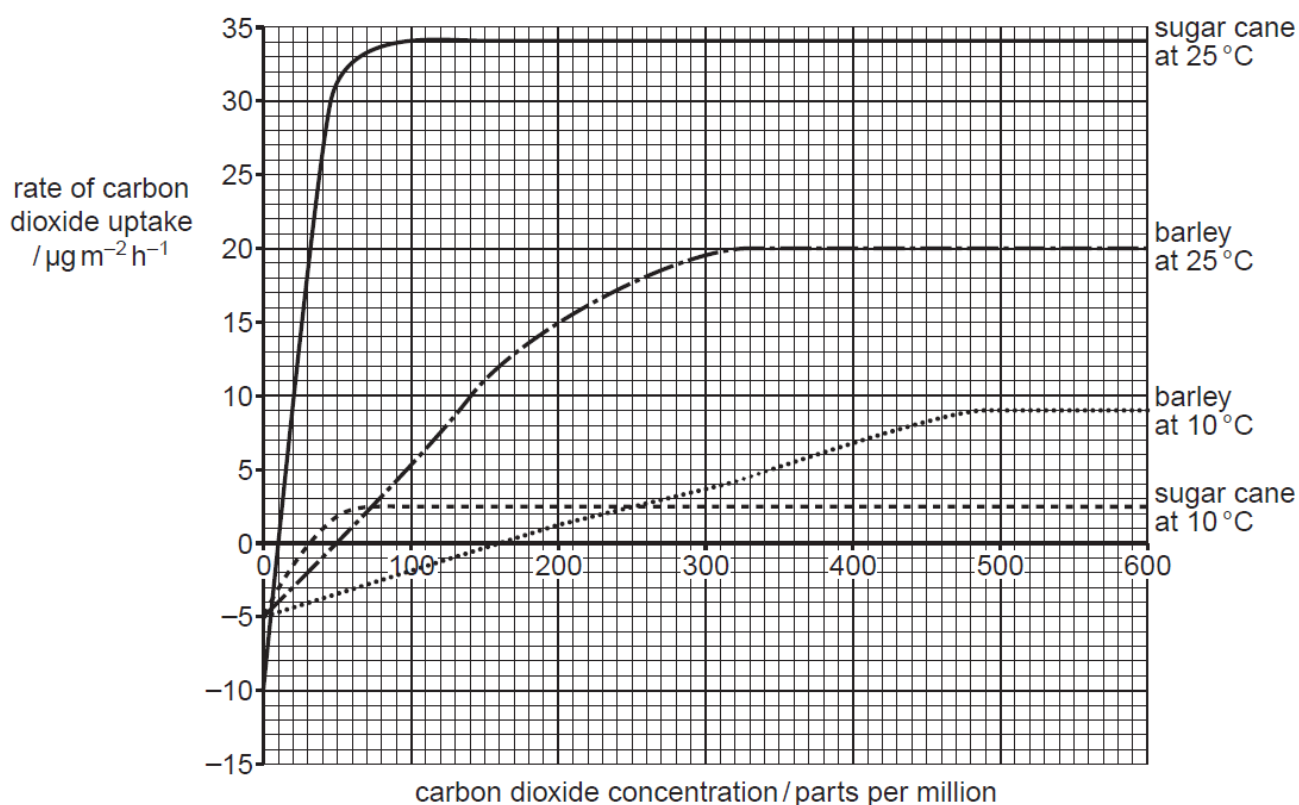


Fig. 2.2

The current carbon dioxide concentration in the atmosphere is more than 400 parts per million and it is likely to increase in the future. It is widely believed that the carbon dioxide concentration of the atmosphere affects the global mean surface temperature which in turn changes rainfall patterns.

Table 2.1 shows other data regarding three C3 and three C4 plants which are important crops.

Table 2.1

crop	mass of water absorbed per gram dry mass produced/ g
rice	682
potato	575
wheat	542
maize	350
millet	285
sorghum	204

With reference to Fig. 2.2 and Table 2.1,

- (d)** discuss the likely impact of the predicted changes in carbon dioxide concentration, global temperatures and rainfall patterns on the global distribution of C3 and C4 plants.

[4]

Climate change potentially affects the spread of diseases. Fig. 2.3 shows the worldwide distribution of dengue.

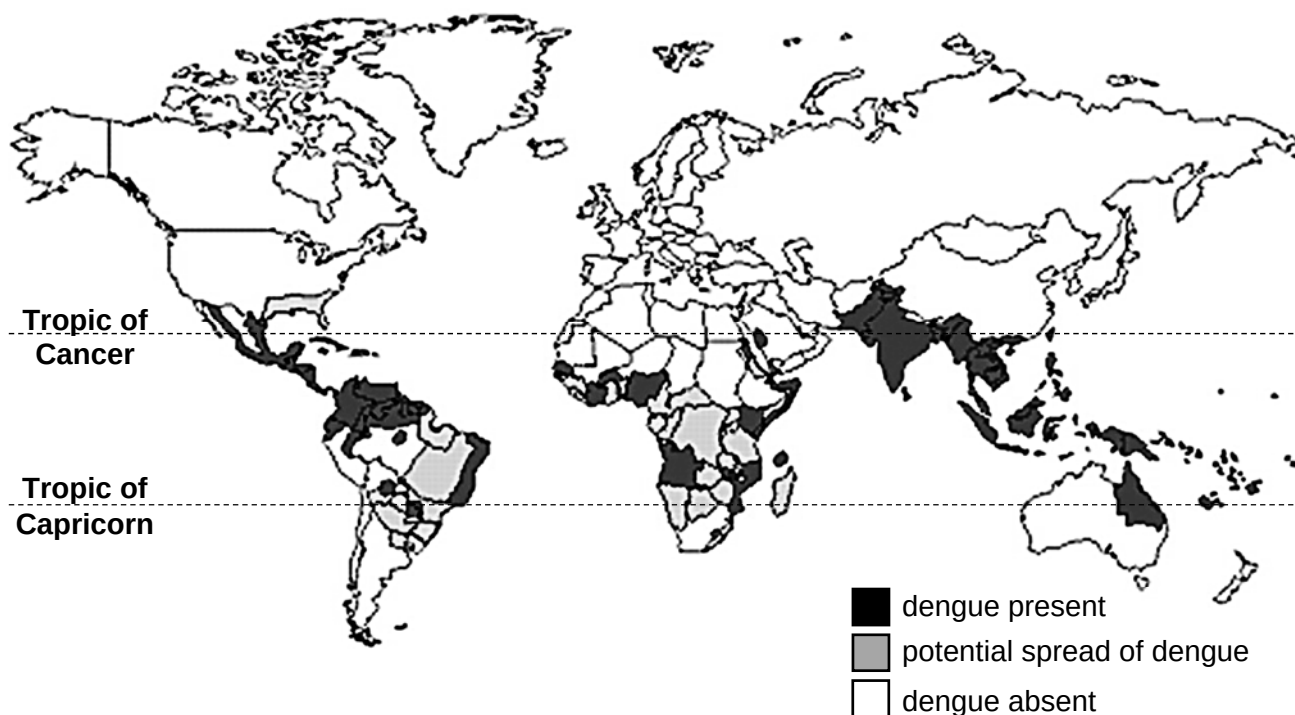


Fig. 2.3

(e) Describe how dengue is transmitted.

[2]

Unlike dengue, influenza is found across the whole world.

(f) Explain why dengue shows the distribution pattern shown in Fig. 2.3, but influenza is found everywhere.

[2]

[Total: 13]

- 3 Hepatitis is the inflammation of the liver and can be caused by a number of different hepatitis viruses. Presently, the only effective vaccines available are for hepatitis A and B.
- (a) Outline the immune response that leads to the production of antibodies after vaccination.

[3]

[5]

This finding has led to pioneering research which produced a successful treatment for hepatitis in mice. The *Fas* gene was silenced using the technique of RNA interference.

This caused the mRNA to be destroyed by enzymes, therefore preventing the gene product from being made. This therapy prevented liver cell death and considerably increased the survival of mice with hepatitis.

- (c) (i)** Describe **one** way in which the function of mRNA differs from that of DNA.

[1]

- (ii) Suggest **one** way in which the structure of siRNA differs from mRNA.

(d) Describe how one strand of the siRNA can bind to the mRNA of the *Fas* gene.

[2]

The technique of RNA interference has also been used to slow down replication of HIV (Human Immunodeficiency Virus) *in vitro*. This is an important breakthrough in the treatment of AIDS as many countries are hit by the epidemic.

The siRNA is attached to a carrier molecule which binds to HIV protein on the plasma membrane of infected cell. This allows carrier with siRNA to enter human cell.

siRNA sequences that match the RNA genome of HIV can be used to trigger destruction of this RNA, preventing HIV from multiplying.

(e) The siRNA would **only** affect gene expression in cells infected with HIV. Suggest **one** reason why.

[1]

Another approach is to use RNA interference to silence genes for cell surface receptors, such as the CD4 and CCR5 molecules on human white blood cells.

If these genes are not expressed, HIV cannot bind to and infect the white blood cells. Table 3.1 summarizes some information regarding the two cell surface receptors used by HIV to bind to and infect white blood cells.

Table 3.1

	cell surface receptor	
	CD4	CCR5
Type of cell with this receptor	T lymphocyte white blood cells which divide by mitosis	Macrophage cells which are long-lived and do not undergo mitosis

Experiments have been carried out where,

- siRNAs matching the CD4 mRNA were introduced into test tube populations of T lymphocytes;
- siRNAs matching the CCR5 mRNA were introduced into test tube populations of macrophages.

In both cases HIV was present and the presence of the siRNAs reduced its replication.

- (f) Using Table 3.1, suggest with reasons which of the two test tube experiments would have a greater reduction in HIV replication.

[2]

Antibiotics are prescribed to people who have HIV/AIDS for the treatment of secondary infections such as bacterial infections.

- (g) Describe the mode of action of antibiotics, such as penicillin, on bacteria.

[3]

- (h) Explain why antibiotics are prescribed to treat secondary infections, but not HIV infection.

 [2]

Antibiotic resistance could develop and the genes for antibiotic resistance could be transmitted between bacteria. Table 3.2 shows features of gene transmission.

Table 3.2

Statement	Vertical	Horizontal
Gene is replicated		
Gene can be passed to other species of bacteria		
Involves conjugation		

- (i) Complete Table 3.2 by putting a tick in the box if the statement is correct for vertical or horizontal gene transmission. [1]

Apart from the devastating effects of HIV, in 2014, parts of West Africa were hit by an epidemic of Ebola fever. Most people who caught the disease died.

Scientist attempted to genetically synthesize an antibiotic as a possible drug to target the Ebola glycoprotein.

This drug was **only** used to treat two Americans who had been working as medics in Africa. Its use was controversial because the drug had not been tested on humans. At the time there were only a few doses of the drug available.

- (j) (i) Suggest a reason why the decision was made to use the drug, even though it had not been tested.

 [1]

- (ii) Apart from the fact that drug had not been fully tested, give **one** reason why using the drug in the way described could be considered as unethical.

 [1]

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must set out in sections **(a)** and **(b)**, as indicated in the question.

4 (a) Describe the causes of variation and its biological importance and how variations are preserved in a population. [10]

(b) Justify the claim that all living organisms on Earth depend on phosphate. [15]

[Total: 25]

5 (a) Biological specificity is one of the most widespread and characteristic properties of living organisms.

Biological specificity is most pronounced and best understood at the cellular and molecular levels of organization. Using named examples, explain the importance of shapes fitting together in cells and organisms. [13]

(b) Describe the effects of different types of mutations on the proteins of eukaryotes. [12]

[Total: 25]

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